

Vikas U Rao

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EDUCATION

Bangalore Institute of Technology

Master of Computer Applications — CGPA: 8.2

Bengaluru, India

Nov. 2025 – Present

EXPERIENCE

Pranava MedLegal Solutions (India) Pvt. Ltd.

Software Intern

Bengaluru, India

Aug. 2025 – Dec. 2025

- Developed a **Patient Demographics Extractor** achieving around **90% accuracy** by identifying and structuring key details from unstructured medical text.
- Contributing to an internal **automation framework** for extracting structured data from PDFs and automating medical worksheet processing.
- Developing an **evaluation framework and pipeline** to automate testing and validation of extraction stages using an **LLM-as-a-Judge** methodology.
- Implemented modules for **PDF-to-image conversion**, **document classification**, and **case aggregation** to streamline data workflows.

PROJECTS

PII Anonymization Gateway | *Python, FastAPI, Microsoft Presidio, spaCy, Gradio, Redis* | [GitHub](#) | [Live Demo](#)

- Engineered a production-grade gateway to detect and anonymize PII before processing sensitive data with LLMs like Gemini and ChatGPT, ensuring HIPAA compliance and data privacy.
- Integrated **Microsoft Presidio** and **spaCy NLP** models to automatically detect 15+ PII types including names, emails, SSN, credit cards, and Indian identifiers (PAN, Aadhaar, GST, Passport).
- Built a **FastAPI** backend with Redis-based session management for real-time PII mapping and stateful de-anonymization of LLM responses.
- Designed an interactive **Gradio** interface demonstrating the complete anonymization → LLM processing → de-anonymization workflow for sensitive data protection.

AI Thumbnail & Metadata Generator | *Python, OpenRouter API, Hugging Face Inference API* | [GitHub](#) | [Live Demo](#)

- Generates YouTube thumbnails and metadata using AI models for automated, high-quality content creation.
- Integrates OpenRouter for text generation and Hugging Face for image generation, supporting multiple styles and overlays.
- Features a modular Python architecture with separate files for UI, API utilities, metadata, image processing, and orchestration.
- Provides a user-friendly Gradio web interface with API key management, real-time generation, and easy export of results.

Amazon Review Sentiment Analyzer | *JavaScript (MV3), Chrome Extensions API, ONNX Runtime Web* | [GitHub](#)

- Built a Manifest V3 Chrome extension performing fully client-side sentiment analysis of Amazon product reviews (no external API calls).
- Implemented resilient multi-selector DOM scraping via `chrome.scripting.executeScript` to adapt to evolving Amazon page structures.
- Integrated cached **Transformers.js** DistilBERT pipeline with review capping + timing metrics for responsive UX and transparency.
- Delivered interactive popup UI showing POSITIVE vs NEGATIVE counts, real-time loading feedback, and robust error handling.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, SQL, HTML/CSS

Frameworks & Tools: Flask, FastAPI, Streamlit, Gradio, Git, Bitbucket, GitHub, VS Code, PyCharm, Google Cloud Platform

Core Competencies: REST API Development, Automation Pipelines, Data Processing, Model Evaluation and LLM Integration, Software Testing